

4	(a)	(i)	Current in R_4 or $R_1 = 0.30 + 0.30 = 0.60 \text{ A}$	C1
			$R = V / I$	M1
			$= 2.4 / 0.60$	
			$= 4.0 \Omega$	A0
		or		
			p.d. across R_3 or $R_2 = 2.4 / 2 = 1.2 \text{ V}$	C1
			$R = V / I$	M1
			$= 1.2 / 0.30$	
			$= 4.0 \Omega$	A0
	(ii)		$E = 2.4 + 2.4 + 1.2$	C1
			$= 6.0 \text{ V}$	A1
		or		
			total resistance $= 10 \Omega$	C1
			$E = 10 \times 0.60$	
			$= 6.0 \text{ V}$	A1
	(b)		total resistance increases	B1
			current decreases (in battery) so total power decreases	B1

5	(a)	(i)	It is to increase the magnetic flux linkage between the coils	B1
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